

```
[1]: import pandas as pd
import numpy as np

[2]: print(np.nan)
print('It's used to print 'Not a number?')

nan
It's used to print 'Not a number'

[3]: print(pd.NaT)
print('It's used to print 'Not a Timestamp')

NaT
It's used to print 'Not a Timestamp'

[4]: np.nan == np.nan

[4]: False

[5]: np.nan is np.nan

[5]: True

[6]: myvar is np.nan
myvar is np.nan

[6]: True

[38]: df = pd.read_csv('C:/Machine Learning/Machine Learning/Data Analysis/03-Pandas/movie_scores.csv')

[8]: df

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0         Tom     Hanks   63.0   m             8.0             10.0
1         NaN       NaN   NaN   NaN             NaN             NaN
2      Hugh   Jackman   51.0   m             NaN             NaN
3      Oprah   Winfrey   66.0   f             6.0             8.0
4      Emma     Stone   31.0   f             7.0             9.0
```

Let's see how to check and select the NaN values

```
[9]: df.isnull()

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0      False      False  False  False           False           False
1       True       True   True   True           True            True
2      False      False  False  False           True            True
3      False      False  False  False           False           False
4      False      False  False  False           False           False

[10]: df.notnull()

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0       True       True   True   True           True            True
1      False      False  False  False           False           False
2       True       True   True   True           False           False
3       True       True   True   True           True            True
4       True       True   True   True           True            True

Check for rows where NaN values are not present in pre_movie_score and post_movie_score

[11]: df[df['pre_movie_score'].notnull() & df['first_name'].notnull()]

   first_name last_name  age  sex  pre_movie_score  post_movie_score
2      Hugh   Jackman   51.0   m             NaN             NaN
```

Keep the data

Drop the data

Fill the data

Keep the data:

```
[13]: df

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0         Tom     Hanks   63.0   m             8.0             10.0
1         NaN       NaN   NaN   NaN             NaN             NaN
2      Hugh   Jackman   51.0   m             NaN             NaN
3      Oprah   Winfrey   66.0   f             6.0             8.0
4      Emma     Stone   31.0   f             7.0             9.0

Drop the data:

[14]: help(df.dropna)

Help on method dropna in module pandas.core.frame:

dropna(
    *,
    axis: Axis = 0,
    how: AnyAll | lib.NoDefault = <no default>,
    thresh: int | lib.NoDefault = <no default>,
    subset: IndexLabel | AnyArrayLike | None = None,
    inplace: bool = False,
    ignore_index: bool = False
) -> DataFrame
    None method of pandas.DataFrame instance
    Remove missing values.

    See the ref: 'User Guide (missing data)' for more on which values are
    considered missing, and how to work with missing data.

    Parameters
    -----
    axis : {0 or 'index', 1 or 'columns'}, default 0
        Determine if rows or columns which contain missing values are
        removed.
        * 0, or 'index': Drop rows which contain missing values.
        * 1, or 'columns': Drop columns which contain missing value.
        Only a single axis is allowed.

    how : {'any', 'all'}, default 'any'
        Determine if row or column is removed from DataFrame, when we have
        at least one NA or all NA.
        * 'any': If any NA values are present, drop that row or column.
        * 'all': If all values are NA, drop that row or column.

    thresh : int, optional
        Require that many non-NA values. Cannot be combined with how.
    subset : column label or iterable of labels, optional
        Labels along other axis to consider, e.g. if you are dropping rows
        these would be a list of columns to include.
    inplace : bool, default False
        Whether to modify the DataFrame rather than creating a new one.
    ignore_index : bool, default "False"
        If "True", the resulting axis will be labeled 0, 1, ..., n - 1.
        .. versionadded:: 2.0.0

    Returns
    -----
    DataFrame or None
        DataFrame with NA entries dropped from it or None if "inplace=True".

    See Also
    -----
    DataFrame.isna: Indicate missing values.
    DataFrame.notna: Indicate existing (non-missing) values.
    DataFrame.fillna: Replace missing values.
    Series.dropna: Drop missing values.
    Index.dropna: Drop missing indices.

    Examples
    -----
    >>> df = pd.DataFrame(
    ...     {
    ...         "name": ["Alfred", "Batman", "Catwoman"],
    ...         "toy": [np.nan, "Batmobile", "Bullship"],
    ...         "born": [pd.NaT, pd.Timestamp("1940-04-25"), pd.NaT],
    ...     }
    ... )
    >>> df
       name      toy      born
0  Alfred      NaN      NaT
1  Batman  Batmobile  1940-04-25
2  Catwoman  Bullship      NaT

Drop the rows where at least one element is missing.

>>> df.dropna()
       name      toy      born
1  Batman  Batmobile  1940-04-25

Drop the columns where at least one element is missing.

>>> df.dropna(axis="columns")
       name
0  Alfred
1  Batman
2  Catwoman

Drop the rows where all elements are missing.

>>> df.dropna(how="all")
       name      toy      born
0  Alfred      NaN      NaT
1  Batman  Batmobile  1940-04-25
2  Catwoman  Bullship      NaT

Keep only the rows with at least 2 non-NA values.

>>> df.dropna(thresh=2)
       name      toy      born
1  Batman  Batmobile  1940-04-25
2  Catwoman  Bullship      NaT

Define in which columns to look for missing values.

>>> df.dropna(subset=["name", "toy"])
       name      toy      born
1  Batman  Batmobile  1940-04-25
2  Catwoman  Bullship      NaT
```

```
[15]: df.dropna()

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0         Tom     Hanks   63.0   m             8.0             10.0
3      Oprah   Winfrey   66.0   f             6.0             8.0
4      Emma     Stone   31.0   f             7.0             9.0

But now we want to remove only that entire NaN row.

[16]: df.dropna(thresh=1)

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0         Tom     Hanks   63.0   m             8.0             10.0
1         NaN       NaN   NaN   NaN             NaN             NaN
3      Oprah   Winfrey   66.0   f             6.0             8.0
4      Emma     Stone   31.0   f             7.0             9.0
```

It means, go ahead and drop all the rows with NaN unless you encounter one not a NaN value

```
[17]: df.dropna(thresh=5)

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0         Tom     Hanks   63.0   m             8.0             10.0
3      Oprah   Winfrey   66.0   f             6.0             8.0
4      Emma     Stone   31.0   f             7.0             9.0

Here, Hugh is gone because it doesn't have 5 non-null values

[18]: df.dropna(axis=1)

0
1
2
3
4

[19]: df.dropna(axis = 0)

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0         Tom     Hanks   63.0   m             8.0             10.0
3      Oprah   Winfrey   66.0   f             6.0             8.0
4      Emma     Stone   31.0   f             7.0             9.0
```

Subset argument:

```
[20]: df.dropna(subset = ['last_name'])

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0         Tom     Hanks   63.0   m             8.0             10.0
2      Hugh   Jackman   51.0   m             NaN             NaN
3      Oprah   Winfrey   66.0   f             6.0             8.0
4      Emma     Stone   31.0   f             7.0             9.0

Here, there's only 1 row with missing last name, so that got removed

Fill the data:

[21]: help(df.fillna)

Help on method fillna in module pandas.core.generic:

fillna(
    value: Hashable | Mapping | Series | DataFrame,
    *,
    axis: Axis | None = None,
    inplace: bool = False,
    limit: int | None = None
) -> Self
    Self method of pandas.DataFrame instance
    Fill NA/NaN values with 'value'.

    Parameters
    -----
    value : scalar, dict, Series, or DataFrame
        Value to use to fill holes (e.g. 0), alternately a
        dict/Series/DataFrame of values specifying which value to use for
        each index (for a Series) or column (for a DataFrame). Values not
        in the dict/Series/DataFrame will not be filled. This value cannot
        be a list.
    axis : {0 or 'index', 1 or 'columns'}, 1 or 'columns' for DataFrame
        Axis along which to fill missing values. For 'Series'
        this parameter is unused and defaults to 0.
    inplace : bool, default False
        If True, fill in-place. Note: this will modify any
        other views on this object (e.g., a no-copy slice for a column in a
        DataFrame).
    limit : int, default None
        This is the maximum number of entries along the entire axis
        where NAs will be filled. Must be greater than 0 if not None.

    Returns
    -----
    Series/DataFrame
        Object with missing values filled.

    See Also
    -----
    ffill : Fill values by propagating the last valid observation to next valid.
    bfill : Fill values by using the next valid observation to fill the gap.
    interpolate : Fill NA values using interpolation.
    reindex : Conform object to new index.
    asfreq : Convert TimeSeries to specified frequency.

    Notes
    -----
    For non-object dtype, "value=None" will use the NA value of the dtype.
    See more details in the ref: Filling missing data (using data.fillna)
    section.

    Examples
    -----
    >>> df = pd.DataFrame(
    ...     [
    ...         [np.nan, 2, np.nan, 0],
    ...         [3, 4, np.nan, 1],
    ...         [np.nan, np.nan, np.nan, np.nan],
    ...         [np.nan, 3, np.nan, 4],
    ...     ],
    ...     columns=list("ABCD"),
    ... )
    >>> df
       A  B  C  D
0  NaN  2  0  0
1  3.0  4  0  1.0
2  NaN  3  NaN  NaN
3  NaN  3  0  4.0

Replace all NaN elements with 0s.

>>> df.fillna(0)
       A  B  C  D
0  0.0  2  0  0
1  3.0  4  0  1.0
2  0.0  3  0  NaN
3  0.0  3  0  4.0

Replace all NaN elements in column 'A', 'B', 'C', and 'D', with 0, 1, 2, and 3 respectively.

>>> values = {"A": 0, "B": 1, "C": 2, "D": 3}
>>> df.fillna(value=values)
       A  B  C  D
0  0.0  2  0  0
1  3.0  4  0  1.0
2  0.0  3  0  NaN
3  0.0  3  0  4.0

Only replace the first NaN element.

>>> df.fillna(value=values, limit=1)
       A  B  C  D
0  0.0  2  0  0
1  3.0  4  0  1.0
2  NaN  3  NaN  NaN
3  NaN  3  0  4.0

When filling using a DataFrame, replacement happens along
the same column names and same indices

>>> df2 = pd.DataFrame(np.zeros((4, 4)), columns=list("ABCE"))
>>> df.fillna(df2)
       A         B         C         E
0  0.0  2.0  0.0  0.0
1  3.0  4.0  0.0  1.0
2  0.0  3.0  0.0  NaN
3  0.0  3.0  0.0  4.0

Note that column D is not affected since it is not present in df2.
```

```
[22]: df.fillna("New Value")

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0         Tom     Hanks   63.0   m             8.0             10.0
1  New Value  New Value  New Value  New Value  New Value  New Value
2      Hugh   Jackman   51.0   m             NaN             NaN
3      Oprah   Winfrey   66.0   f             6.0             8.0
4      Emma     Stone   31.0   f             7.0             9.0
```

If we fill with a string, what Pandas does is, it converts all the values, such as 63 here as a string instead of its original datatype.

```
[23]: df['pre_movie_score'] = df['pre_movie_score'].fillna(0)

[24]: df

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0         Tom     Hanks   63.0   m             8.0             10.0
1         NaN       NaN   NaN   NaN             0.0             NaN
2      Hugh   Jackman   51.0   m             0.0             NaN
3      Oprah   Winfrey   66.0   f             6.0             8.0
4      Emma     Stone   31.0   f             7.0             9.0
```

Let's try filling them with mean

```
[25]: mean = df['pre_movie_score'].mean()

mean

[25]: np.float64(4.2)

[26]: df['pre_movie_score'].fillna(df['pre_movie_score'].mean())

a
a.a
[28]: a
0      0.0
1      0.0
2      0.0
3      6.0
4      7.0
Name: pre_movie_score, dtype: float64

[29]: df

   first_name last_name  age  sex  pre_movie_score  post_movie_score
0         Tom     Hanks   63.0   m             8.0             10.0
1         NaN       NaN   NaN   NaN             0.0             NaN
2      Hugh   Jackman   51.0   m             0.0             NaN
3      Oprah   Winfrey   66.0   f             6.0             8.0
4      Emma     Stone   31.0   f             7.0             9.0
```

```
[31]: df.fillna(df.mean())

Traceback (most recent call last)
<ipy>:1: TypeError: Traceback (most recent call last)
Call 100/100: TypeError:
>>> df.fillna(df.mean())

File c:\Users\paul\Anaconda3\local\Programs\Python\Python38\1\lib\site-packages\pandas\core\frame.py:1487, in DataFrame.mean(self, axis, skipna, numeric_only, **kwargs)
>>> df.fillna(df.mean())
1487:         func = self._mgr.reduce(func, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1488:         return self._stat_function(
1489:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1490:         return self._stat_function(
1491:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1492:         return self._stat_function(
1493:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1494:         return self._stat_function(
1495:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1496:         return self._stat_function(
1497:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1498:         return self._stat_function(
1499:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1501:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1503:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1505:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1507:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1509:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1511:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1553:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1557:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1559:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1561:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1565:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1571:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1617:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1623:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1624:         return self._stat_function(
1625:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1626:         return self._stat_function(
1627:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1629:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1631:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1633:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1634:         return self._stat_function(
1635:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1636:         return self._stat_function(
1637:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1638:         return self._stat_function(
1639:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
1640:         return self._stat_function(
1641:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1643:             func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs)
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1826:         return self._stat_function(
1827:             func, name=name, axis=axis, skipna=skip
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